

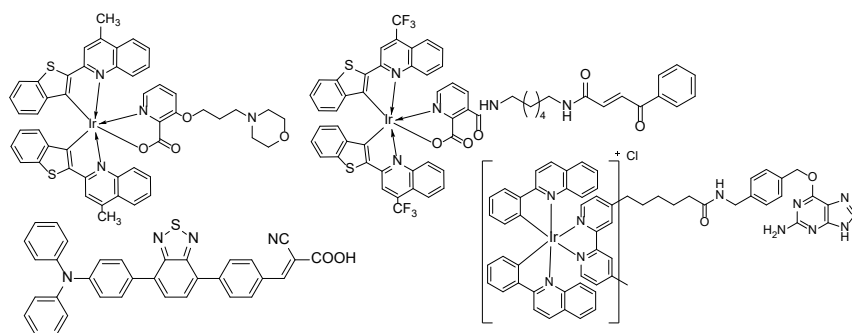
Molecular Design & Synthesis

Organic Synthesis

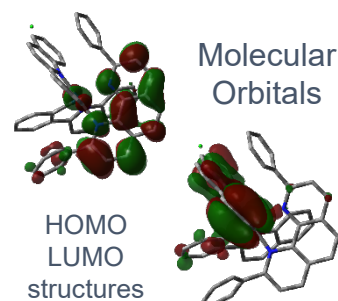
Photosensitizers

DFT Calculation

Photoresponsive organic & organometallic compounds



Material design with DFT



Characterization

NMR

UV-Vis

Steady-state/Transient PL

HPLC

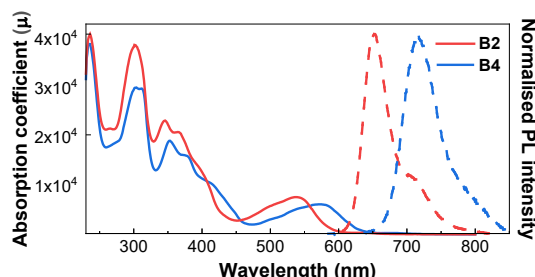
LC-MS

CV

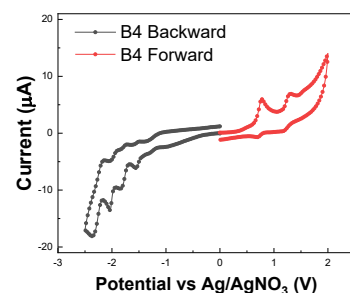
Chemical analysis

- NMR (^1H , ^{13}C)
- Liquid Chromatography
- Mass spectrometry
- Melting Point
- Characterization of photoreactive compounds and their rxn products

Spectroscopic analysis



Electrochemical analysis



In vitro experiments

Cell culture

SDS/Native PAGE

IF

WB

Transfection

Cytotoxicity assays

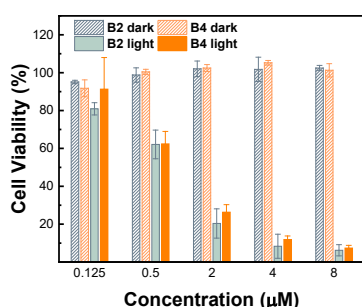
ROS assays

Confocal microscopy

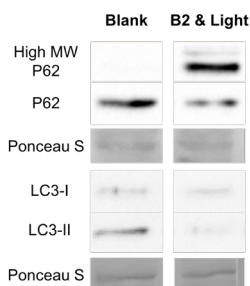
Fixed/Live cells

Tissue

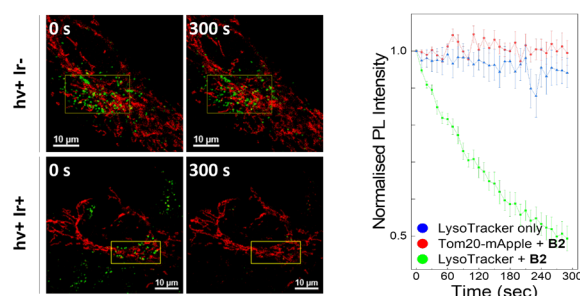
Cell-based assays



Immunoblots



Microscopic analysis



Proteomics

Global proteomics

In-gel

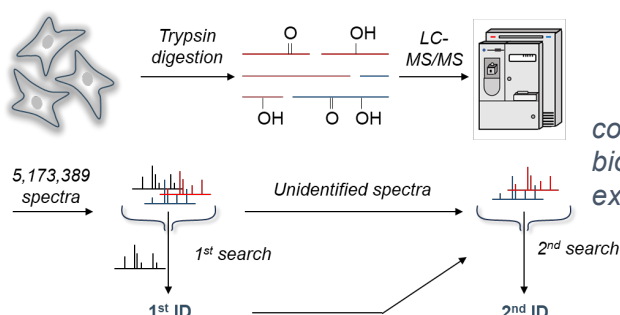
On-filter

Data analysis

Data visualization

PTMs

Proteomic sampling and analysis



collab. with
bioinformatics
experts

Proteome visualization

